CCIT Report No. 307

Coding Schemes for Multislot Messages in Multichannel ALOHA with Deadlines

Dror Baron and Yitzhak Birk

ABSTRACT

Slotted multichannel ALOHA is the access scheme of choice for short messages and to reserve channels for longer ones in mnay statellite-based networks. This paper proposes schemes for increasing capacity of multichannel Slotted ALOHA subject to a user-specified deadline and a (small) permissible probability of exceeding it, thereby jointly capturing the user requirements and the system owner's desires. The focus is on short yet multislot messages. A key idea is to achieve a low probability of missing the deadline by permitting a large maximum resource expenditure per message, while holding the mean expenditure low in order to minimize "pollution". For a K-slot message, redundant single-slot fragments are constructed using block erasure-correcting codes, such that any K fragments suffice for message reception. With Multiround Coding, an optimized number of fragments are transmitted in each round until K are received or the deadline is reached. Even with very strict constraints, capacities that approach the $1/e$ limit are attained. The Coding-Reservation scheme raises capacity above $1/e$ by allowing the hub, upon receipt of a message fragment, to grant contention-free slots for the remaining required fragments. Both schemes are adapted for use with single-transmitter stations at a small performance penalty in most cases. Finally, because capacity is maximized by minimizing the mean per-message transmission resources, the transmission scheme is also power efficient.